Canada Optometry Scraper — Nova Scotia (Pilot)

Collects **optician stores and optometry practices** across Nova Scotia, Canada and exports a clean **CSV** and **Excel** file. Built to scale to every province and territory by changing one bounding box and city list.

Why this approach (read before running)

"Crawl the internet" sounds like scraping Google search or Yellow Pages directly. Doing that gets your IP blocked, breaks every time a page layout changes, and is often against those sites' Terms of Service — so it fails a paid test fast and isn't defensible to a client. This project instead uses **structured data sources** that are legal, stable, and built for exactly this:

Source	Cost	API key	Coverage	Legal
OpenStreetMap (Overpass)	Free	None	Good	Open data (ODbL)
Google Places API	~$32 USD / 1000 req (free tier first)	Yes	Best	ToS-compliant

OSM runs immediately with no key. Add a Google key for maximum coverage. The two sources are deduplicated and merged, so richer Google data fills gaps in OSM.

> For a client deliverable, also cross-check against the **Nova Scotia College of Optometrists** public "Find an Optometrist" directory — that's the authoritative registry of *licensed* practices. See the note at the bottom.

Quick start (VS Code)

1. Open this folder in VS Code (`File → Open Folder`).
2. Create a virtual environment and install dependencies:

```
python -m venv .venv
# Windows:
.venv\Scripts\activate
# macOS / Linux:
source .venv/bin/activate
pip install -r requirements.txt
```

3. Run it (free, no key needed):

```
python src/scraper.py
```

4. Find your files in `data/`:

- `nova_scotia_optometry.csv`
- `nova_scotia_optometry.xlsx`

Or just press **F5** in VS Code and pick **"Run scraper (OSM only)"**.

Adding Google Places (recommended for full coverage)

1. Create a Google Cloud project, enable **Places API**, create an API key.
2. Set the key as an environment variable:

```
# Windows (then reopen terminal):
setx GOOGLE_API_KEY "your_key_here"
# macOS / Linux:
export GOOGLE_API_KEY="your_key_here"
```

3. Run with Google enabled:

```
python src/scraper.py --google        # OSM + Google merged
python src/scraper.py --google-only   # Google only
```

Output columns

`name, category, address, city, postal_code, province, phone, website, email, latitude, longitude, source, place_id`

Scaling to all provinces & territories

Each region needs only its bounding box and a city list. To extend, parameterize `NS_BBOX` / `NS_CITIES` per province (e.g. a `provinces.py` dict) and loop. The collect → dedupe → write pipeline is unchanged.

Manual authoritative source (do this for the client)

The **Nova Scotia College of Optometrists** publishes a public directory of licensed optometrists. It has no open API, so it should be collected manually or with explicit permission, and used as the source of truth to validate the scraped list. This keeps the deliverable accurate and defensible.

Notes & limits

- Overpass public endpoints rate-limit; the script tries several mirrors.
- Respect each source's Terms of Service and licensing (OSM = ODbL attribution).
- Verify a sample of records before delivering to a client.